

Deepak Mallampati

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Kent, OH (open to SF relocation) -- F-1 OPT, open to sponsorship

PROFESSIONAL SUMMARY

Applied AI engineer (2.5+ years) who ships AI-powered internal tools and automations end-to-end — operator pain to production in days. Solo-shipped **Manga Lens** Chrome extension (4 AI vision providers, 29 sites, Chrome Web Store) and **Agentic Healthcare Claims** multi-agent pipeline with schema-validated JSON contracts to eliminate cascading hallucinations. Production track record: ~35% RAG precision lift, >30% hallucination reduction, 15-20% recall gain on high-risk patient categories, ~30% fewer post-deployment defects in HIPAA-conscious healthcare. Comfortable embedding with non-technical operators, translating their friction into shipped automations.

CORE COMPETENCIES

Internal Tools & Workflow Automation . Forward-Deployed Delivery . Agentic LLM Systems . Production RAG . Schema-Validated Multi-Agent Pipelines . Python / TypeScript / FastAPI . Operator-Embedded Engineering . Regulated-Domain Governance (HIPAA → CFPB) . Solo Ship Discipline . End-to-End Ownership

EXPERIENCE

Progress Solutions Inc.

Jul 2025 -- Present

Jr. AI/ML Engineer Trainee -- USA | Healthcare Technology, AI Consulting

- Built **agentic LLM workflows** for multi-step healthcare queries (eligibility checks, care navigation, documentation clarification) with structured reasoning, tool discipline, and grounding rules; improved agent stability ~25% and reduced hallucination rate >30%.
- Built **RAG systems** for clinical knowledge retrieval; recursive semantic chunking + transformer embeddings improved precision ~35% and reduced irrelevant retrieval >30%; response alignment exceeded 90%.
- Packaged LLM/ML inference as **FastAPI/Flask** RESTful services containerized with Docker; structured logging + load simulation cut post-deployment defects ~30%.
- Authored stakeholder-facing system-limitation docs and audit trails for non-technical clinical operators — enabled adoption in HIPAA-conscious environment.
- Shipped predictive ML (scikit-learn, XGBoost) for patient no-show and care engagement; improved recall 15-20% on high-risk while holding precision >90%.

Energy Solutions International (Emerson)

Jun 2022 -- Apr 2023

Database & DevOps Performance Engineer (Intern) -- Hyderabad, India | Enterprise ERP

- Built CI/CD pipelines with Jenkins for the Synthesis Order-to-Cash ERP platform; reduced deployment errors >30%, improved release cycle ~35-40%.
- Designed Grafana observability dashboards for production SQL systems; cut incident recurrence ~25%, improved RCA speed ~30%.

Suvidha Foundation

Jan 2022 -- Mar 2022

Machine Learning Engineer -- Hyderabad, India | Nonprofit Tech, Applied NLP

- Built transformer-based video summarization for 5,000+ sessions; reduced manual review time 60-70%; deployed as Flask API for non-technical users.
- Implemented document Q&A with RAG; reduced hallucinated outputs ~30%.

PROJECTS

Manga Lens shipped -- Chrome Web Store

Solo-built Chrome extension integrating **4 AI vision providers** (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, local Ollama) via provider-agnostic abstraction. Multi-section capture pipeline for tall webtoon panels, 7-day translation cache, 29 site selector configs. Full ship discipline: Manifest V3, service workers, scoped permissions, privacy policy, Chrome Web Store review. **Demonstrates fast solo ship from problem to production.**

TypeScript, Manifest V3, content scripts, multi-provider abstraction, captureVisibleTab

Agentic Healthcare Claims Processing & Fraud Risk Intelligence

Multi-agent pipeline (Intake -> Validation -> Consistency -> Duplicate Detection -> Fraud Risk) with **schema-validated JSON contracts between agents** to prevent cascading hallucinations. RAG-grounded CPT/ICD validation; ANN similarity for duplicate detection; explainable risk scoring with audit-ready reasoning traces. Direct analog to insurance underwriting / brokerage operational AI workflows.

Python, LangChain, vector search, FastAPI, Docker, regulated-domain governance

Dream Decoder -- AI Multimodal Platform

Full-stack FastAPI + React/TypeScript app coordinating multimodal AI APIs (Perplexity Sonar for interpretation, Replicate for visuals). Intermediate structured prompt transformation layers improved contextual alignment **~30%** and first-pass image success **~25-30%**.

FastAPI, React, TypeScript, Vite, multi-provider LLM orchestration

EDUCATION

Master's Degree -- Kent State University, Ohio, USA

Aug 2023 -- May 2025

TECHNICAL SKILLS

Languages & Frameworks: Python, TypeScript, JavaScript, React, FastAPI, Flask, SQL (PostgreSQL, T-SQL, SQLite)

AI/LLM & Agentic: LLM APIs (Claude, GPT-4o, Gemini, Ollama), LangChain, LlamaIndex, RAG, multi-agent workflows, schema contracts, prompt engineering, structured outputs, evaluation pipelines, guardrails, grounding

Internal Tools / Automation: Chrome extensions (Manifest V3), full-stack web apps (FastAPI + React), workflow orchestration, multi-provider API integration

Developer Tooling: Cursor, Claude Code, Docker, CI/CD (Jenkins), Grafana, REST APIs, MLOps/LLMOps, observability/logging

Regulated Domains: HIPAA-conscious governance, de-identification, audit trails, stakeholder-facing system-limitation docs — transferable to CFPB / state insurance compliance

Work Authorization: F-1 OPT -- authorized to work in the United States now; future sponsorship required